PH7002: Concepts in Statistical Mechanics
§3 Mean Field Theory

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Reading

Yeomans sections 4.1.

Very few statistical mechanics model can be solved exactly. Therefore there's a need for approximation. In this chapter, we will study a popular method known as **mean field theory**. This method will give us a quantitative accounting of the properties of the various phases. The mean field method has important strengths, and also weaknesses, which will become more apparent as the course goes on.

Important themes

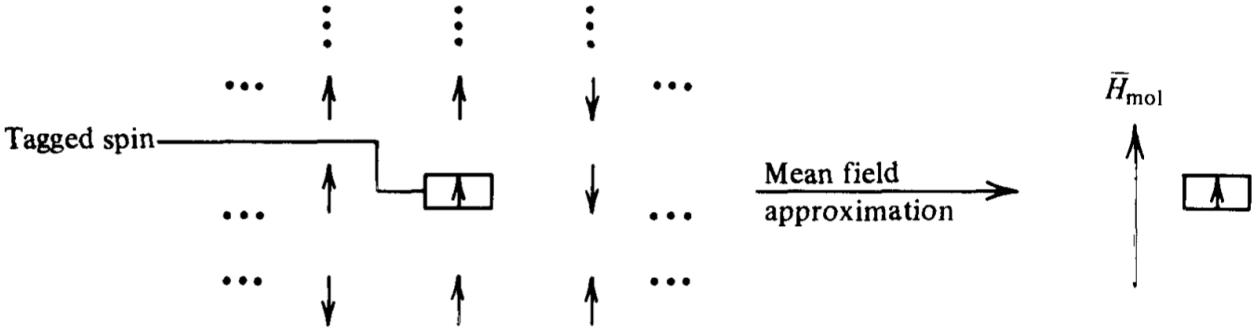
1. Understand basic principles behind mean field theory.
2. Use mean field theory to solve Ising model and other physical systems.
3. Calculate critical exponents in mean field theory.

Ising model (zero field)

Mean field theory is the simplest treatment of an *interacting* statistical mechanical system, apart from the approximation of ignoring the interactions all together. The term "mean field theory" conveys an impression of uniqueness, but this is false: there are many ways to generate mean field theories. To illustrate the method, we will apply it to the Ising magnet.

The idea is to focus on one particular particle (in this case a spin) in the system and assume that the role of the neighboring particles (spins) is to form an average molecular (magnetic) field which acts on the tagged particle (spin). This approach neglects the effects of fluctuations that extend beyond the length scale associated with the primary or tagged lattice cell.

The method includes only those fluctuations that occur within the tagged cell, and since these involve only one particle, the method succeeds at reducing the many body statistical mechanics problem into a few (i.e., one) body problem. Procedures of this type can never be exact, but often they can be very accurate and quite useful. They are least accurate, of course, for systems near critical points where cooperativity in fluctuations extends over large distances.



Let us write the Ising model energy

The force exerted on s_i due to the neighboring spins and the external field, H, is

Therefore, the instantaneous field impinging on spin i is given by

As the neighboring spins fluctuate, H_i fluctuates from its mean

where in the second equality we have noted that $\langle s_i \rangle = \langle s_j \rangle$ for all i and j in the Ising model, and z is the number of nearest neighbors around a given tagged spin. It is the fluctuations of H_i away from its mean that couples the fluctuations of the tagged spin to those in its environment. Indeed, in the mean field approximation which neglects the deviations of H_i from $\langle H_i \rangle$, the force on a tagged spin is independent of the instantaneous configuration of the neighboring spins.

We can compute the average value of a tagged spin, labeled 1 for convenience, in the mean field approximation

where ΔH is the molecular or environmental contribution to the mean field; that is

Performing the sums over the two spin states yields

where $\langle s \rangle$ is the magnetization per particle.

This is a transcendental equation whose solution yields the magnetization predicted by this self-consistent mean field theory. It is "**self-consistent**" in the sense that the mean field which influences the average value of the magnetization depends itself upon the average of the magnetization.

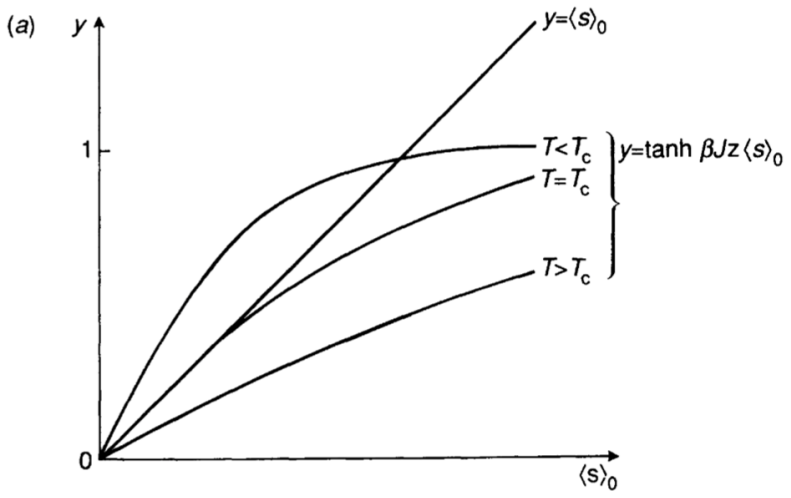
The orientation at every site is just such as to create the same orientation at all other sites.

We can solve graphically. Notice that $h = 0$—no orientation—is always a self-consistent solution.

For $T < T_c$ there are two better solutions.

For $T > T_c$, this is the only solution.

The boundary between the 2 behaviors occurs when the function $y = \langle s \rangle_0$ and $y = \tanh \beta J z \langle s \rangle_0$ are tangent at $\langle s \rangle_0 = 0$.

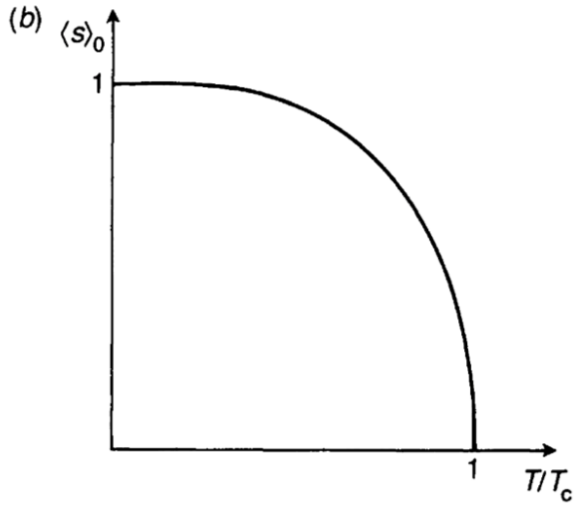


Non-zero solutions for $\langle s \rangle$ exist for $H = 0$ when $\beta J z > 1$. Thus, for a square lattice we predict the critical temperature

Let us expand about $\langle s \rangle_0 = 0$

Tangency occurs when

Following the ferromagnetic solution as T is decreased from T_c, we see that $\langle s \rangle_0$ increases continuously from zero, i.e. phase transition is continuous. As $T \to 0$,



For $T > T_c$, $\langle s \rangle_0 = 0$; for $T < T_c$, but close to T_c, we can solve for $\langle s \rangle_0$ in a power series expansion in $T - T_c$.

The magnetization as a function of temperature is therefore given by

Notice that for one dimension, the mean field theory predicts that order-disorder transition occurs at a finite temperature

But the exact analysis of the one-dimensional Ising model yields

i.e., no spontaneous magnetization at finite temperatures. Thus, the theory is "infinitely wrong" in one dimension. Spontaneous fluctuations destroy the order. But mean field theory, which neglects fluctuations, predicts long range order. In two dimensions, the Onsager solution yields $T_c \approx 2.27J/k_B$, while the theory yields $T_c = 4J/k_B$. In three dimensions, the percentage error is smaller: the correct answer (known from numerical work) is $T_c \approx 4J/k_B$, whereas the theory gives $T_c = 6J/k_B$. In each case, the neglect of fluctuations predicts transitions to an ordered state at temperatures higher than the true transition temperature.

Ising model (external field)

Let us repeat the analysis with non-zero magnetic field $h \neq 0$. Due to external field, Z_2 symmetry is explicitly broken. When $J = 0$, we find that

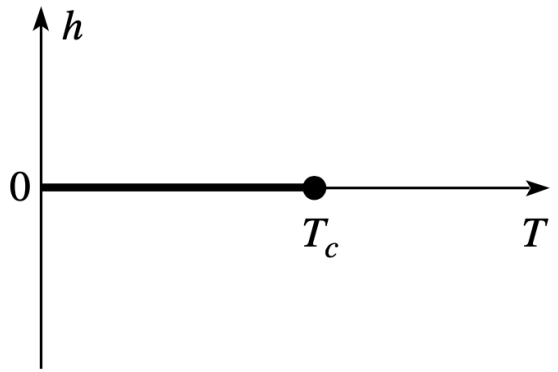
At $h = 0$, we obtain

This is the **Curie-Weiss law**.

For $J \neq 0$, we find

The solution for $h \neq 0$ and small β (large T) evolves smoothly into the favored solution for $h \neq 0$ and large β as β is increased. All of the M vs T contours are smooth except for the contour at $h = 0$.

The phase diagram of h vs T appears as



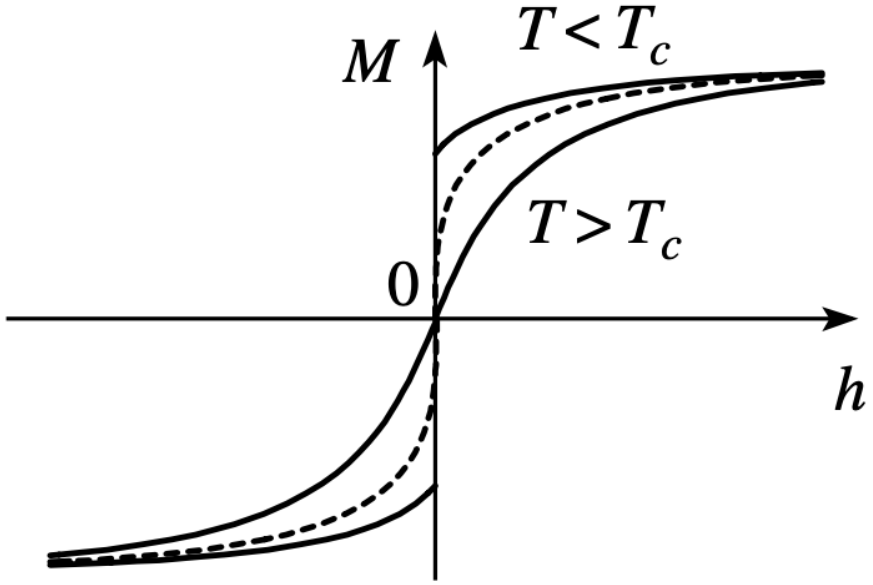
For $T > T_c$, $M = 0$ at $h = 0$, and changes continuously as h is varied from negative to positive values. For $T < T_c$, M takes different values on opposite side of the line $h = 0$. As h is increased from negative to positive values, M changes continuously across this line. The special point $T = T_c$, $h = 0$ where essential non-analytic behavior occurs is the critical point.

Let us calculate the magnetic susceptibility at non-zero field for $T > T_c$. Use

We see that the **Curie-Weiss law** is modified such that **singularity** is now at $T = T_c$. It can also be shown that

as T approaches T_c from below.

The same analysis gives the magnetization as a function of h at $T = T_c$.


M
T < T_c
T > T_c
0
h

To calculate α_{MF}, we will calculate $\langle E \rangle$ and then find $C_v = \partial \langle E \rangle / \partial T$. In mean field approximation,

Let me summarize. The various thermodynamic functions are non-analytic at the point $T = T_c$, $h = 0$ with the singular behavior

Variational Treatment of Mean Field Theory

The Ising model problem is the computation of the Boltzmann weighted sum

The interaction terms, $-Js_is_j$, are responsible for cooperativity, but also for the complexity of computing Z. Let us consider an energy function of the form

A system with this energy function is a model with independent spins, and each spin fluctuates under the influence of a static or mean field. This model presents a simple statistical mechanical problem. The partition function is

and the average of any of the s_i's is

In adopting a mean field description, the static molecular field ΔH must be identified. Previously, we have set

The question we now ask is whether this identification was actually the **optimum** one. In particular, once we adopt the physical picture associated with $E_{\mathrm{MF}}(s_1, \ldots, s_N)$ (i.e., independent spins fluctuating in an effective static field), we can strive to optimize the parameterization of that model. In this case, the mean field model is particularly simple being characterized by only one parameter, ΔH.

To optimize, we consider performing a perturbation theory with the mean field model as the reference. This is done as follows: Let

Then

where the arguments of $E_{\mathrm{MF}}(s_1, \ldots, s_N)$ and $\Delta E_{\mathrm{MF}}(s_1, \ldots, s_N)$ are understood but omitted for notational simplicity. By factoring the summand, we can also write

Now, let us define

as the Boltzmann averaging operation which is weighted with the mean field energy function $E_{\mathrm{MF}}(s_1, \ldots, s_N)$. Then,

This last equation, which is an exact factorization of the partition function, is the starting point for **thermodynamic perturbation theory**.

Consider the **Hölder inequality**:

Gibbs-Bogoliubov-Feynman bound

where F it the true free energy of the system with Hamiltonian $\mathcal{H}$, $\mathcal{H}_0$ is a trial Hamiltonian depending on a variational parameter H_0, F_0 is the corresponding free energy, and $\langle \cdots \rangle_0$ denotes an average taken in the ensemble defined by $\mathcal{H}_0$.

The mean field free energy is then defined by minimizing Φ w.r.t. the variational parameter H_0

This gives the best possible approximation to the true free energy for a given choice of $\mathcal{H}_0$: the inequality means the mean field free energy is always larger than the true free energy. The usual choice of $\mathcal{H}_0$ is a free Hamiltonian - one with no interactions -as this enables explicit calculation.

Let us use the **Gibbs-Bogoliubov-Feynman bound** to optimize the mean field reference system since we can adjust the molecular field ΔH so as to maximize the right-hand side. That is, ΔH is determined by solving the equation

The calculation proceeds as follows: First,

where we have used the fact that since spins are uncorrelated in the mean field model, $\langle s_i s_j \rangle_{\mathrm{MF}} = \langle s_i \rangle_{\mathrm{MF}} \langle s_i \rangle_{\mathrm{MF}}$ for $i \neq j$, and since all spins are the same on the average, $\langle s_i \rangle_{\mathrm{MF}} = \langle s_j \rangle_{\mathrm{MF}}$.

In all, we find

In this 1d Ising model example, we see that the optimal value of ΔH is actually identical to what we have adopted on physical grounds previously. Therefore, we obtain the same **self-consistent mean field equations** studied in that section. Those equations are therefore the **optimum** theory that can be constructed once one adopts the mean field model in which the Ising magnet is pictured as a system of independent spins, each one fluctuating under the influence of an average static environment.

If we compare to the critical exponents, we see that they do not quite agree.

Table 3.1. Universality classes

Universality class	Symmetry of order parameter	α	β	γ	δ	ν	η	Physical examples
2-d Ising	2-component scalar	0 (log)	1/8	7/4	15	1	1/4	some adsorbed mono e.g. H on Fe
3-d Ising	2-component scalar	0.10	0.33	1.24	4.8	0.63	0.04	phase separation, flu order-disorder e.g. β
3-d X-Y	2-dimensional vector	0.01	0.34	1.30	4.8	0.66	0.04	superfluids, supercon
3-d Heisenberg	3-dimensional vector	−0.12	0.36	1.39	4.8	0.71	0.04	isotropic magnets
mean-field		0 (dis.)	1/2	1	3	1/2	0	
2-d Potts, q=3	q-component scalar	1/3	1/9	13/9	14	5/6	4/15	some adsorbed mono
q=4		2/3	1/12	7/6	15	2/3	1/4	e.g. Kr on graphite

Summary

Mean field theory usually gives correct qualitative predictions for the phase diagrams of three-dimensional systems. It provides a feasible approach to complicated problems (often the only one), and it can be used as a starting point for more sophisticated calculations. Whereas many numerical approaches get harder as the dimensionality increase, mean field theory improves. Finally, it highlights the importance of the symmetry of the order parameter.

Extra Notes